



AI Cohort Session: Summary & Architectural Reference

🧠 Deep-Dive on Second Brain Systems, Autonomous HFT Agents & Serverless Knowledge Workflows ✨



1. Multi-Tiered Second Brain (PARA)

- **Scale:** 100,000+ personal and technical documents (~100 GB vault).
- **Architecture:** Organised via *Projects, Areas, Resources, and Archive*.
- **Smart Hybrid Search:** Combines Grep (local fast search), Google Drive (OCR/Images), Qdrant (Vector semantic), and Neo4j (200k+ Entity relationships).
- **Token Efficiency:** Tiered routing drastically reduces query token consumption compared to full-context dumping.



2. Autonomous Prediction Trading

- **Platform Edge:** Binary prediction markets (Polymarket) over high-volatility crypto day trading.
- **Latency Advantage:** Bots ingest direct weather station telemetry before public exchanges update (~100ms delta).
- **Infrastructure:** 24/7 autonomous agents (Hermes) hosted on low-latency Ireland VPS close to server hubs, controllable via Telegram.
- **Discipline:** Autonomous verification of mathematical "edge/alpha" before entering market orders.



3. Serverless Staging & Cloudflare

- **Automated Pipeline:** Ingests raw chat logs into an Obsidian staging vault that auto-sorts topics and schedules reminders.
- **Deployment:** Served directly on **Cloudflare Workers** for zero-latency web accessibility.
- **Security Layer:** Guarded using Cloudflare Zero Trust access policies to protect proprietary notes and formulas.



4. Enterprise & Research Ingestion

- **Longevity Ecosystem:** Ingesting global science papers and longevity market ecosystem data into local Obsidian/Claude pipelines.
- **Email CRM:** Extracting historical contact conversations to build automated background CRM workflows without manual entry.



Technology Stack & Architecture Reference

Layer / Domain	Tool / Service	Role & Architecture Purpose
ORCHESTRATION	LangChain & LangGraph	Frontal cortex orchestrator; routes queries across databases and triggers autonomous background tools.
AUTONOMOUS AGENT	Hermes Agent	24/7 background executor; runs tasks, handles notifications, and interfaces through Telegram.
KNOWLEDGE VAULT	Obsidian Vault	Markdown-based knowledge base for PARA project tracking, daily workflows, and local dashboard UI.
GRAPH DATABASE	Neo4j (Fly.io)	Maps 200,000+ relationships across entities, projects, corporate contacts, and locations.
VECTOR DATABASE	Qdrant (Fly.io)	Semantic similarity vector search for unstructured documents, query embeddings, and natural search.
RELATIONAL DB	EDB Postgres	Maintains relational data consistency, transactional integrity, and verified record schemas.
DOCUMENT ARCHIVE	Google Shared Drives	Large-scale OCR, full-text document indexing, and native image recognition across 5 TB data tiers.
DATA SYNC	Rclone (PAP Pipeline)	Automated <i>Pull-Push-Archive</i> pipeline syncing local nodes, Proxmox hypervisors, and cloud databases.
EDGE & HOSTING	Cloudflare Workers	

Layer / Domain	Tool / Service	Role & Architecture Purpose
		Serverless web dashboard hosting protected by Cloudflare Zero Trust authentication layers.
MARKETPLACE	Polymarket	High-frequency binary prediction platform executed automatically via algorithmic bots.

 Key Principles & Strategic Takeaways

- **Focus on Strategy Over Execution:** Since agents handle code generation, data sorting, and trade execution, the human advantage lies in designing edge hypotheses, risk limits, and architectural setups.
- **Tiered Search Over Big Contexts:** Avoid passing entire raw knowledge bases into LLM contexts. Use tiered retrieval (Grep → Full-Text → Semantic → Graph) to keep latency low and token costs near zero.
- **Start Simple, Grow Iteratively:** Begin with standard local Obsidian or Google Drive setups before scaling to multi-database clusters and background daemon workflows.

 Community Events & Next Steps

 Venture Café London & Cambridge

Weekly Thursday community meetups in September at Cambridge Glass House & London Triton Square (Anthropic HQ).

 Hands-On Database Workshop

Upcoming cohort session focusing on configuring LangChain, LangGraph, and Google Drive integrations step-by-step.